Quantitative Methods 1B

SUSTAINABLE DEVELOPMENT GOALS

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Group Project.

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Effect of 2015 Sustainable Development Goals on Global Low-Carbon Usage

Abstract

This report investigates the effectiveness of the 2015 Sustainable Development Goals (SDGs) on the adoption of low-carbon energy. Using data from 77 countries between 1992 and 2024, our analysis investigates shifts in low-carbon energy shares pre- and post-2015 and evaluates whether national energy policies influence LCE adoption using linear regression and panel fixed effects models whilst controlling for GDP per capita. The results show that the SDGs are associated with a statistically significant increase in shares of low-carbon energy, with post-2015 growth rates rising by an average of 0.45 percentage points ($p < 0.001$) – an effect stronger than that of GDP per capita – and that countries with more energy policies also demonstrate significantly higher LCE shares, even after accounting for time effects and country characteristics. Our findings indicate that global frameworks and national policy commitments play a significant role in advancing low-carbon transitions, making it essential – given the urgency of the climate crisis – to assess the effectiveness of current international and national efforts in order to identify the strategies that genuinely drive progress towards a zero-carbon future.

Introduction

Increasing global concerns on climate change have prompted governments, international organisations and individuals to seek more sustainable energy sources. In 2015, 193 UN member states adopted the Sustainable Development Goals (SDGs), including SDG 7 – Affordable and Clean Energy – which calls for significant global efforts to increase renewable and low-carbon energy shares (United Nations, 2015), crucial in mitigating climate change and facilitating environmental sustainability. SDG 7 is among those goals which – although uneven – have seen the most progress, particularly in European countries, and much less so in East Asian and African countries; the extent of the SDGs' influence in renewable energy transitions remains unclear. While global frameworks set a precedent for government targets, national policies are significant catalysts in transitioning away from fossil fuels. Owing to several factors – particularly level of development and political landscape – both the number and strength of energy policies vary greatly between countries, and therefore so does the rate of adoption of and investment into low-carbon energy.

This report investigates two research questions intersecting global and national policy to assess the impact of SDGs. Firstly, we assessed if there was evidence that the introduction of SDG 7 accelerated the adoption of low-carbon energy globally. Secondly, we investigated whether countries with a greater number of energy policies exhibited higher shares of low-carbon energy consumption. Given the urgency of climate change and the inadequate progress of many SDGs, our analysis aims to set about quantifying the impact of the SDGs and national policy counts on the global transition to renewable energy, thereby addressing a gap in current literature, and providing insight into whether additional efforts might be needed to reach the 2030 goals. The rest of the paper presents the data and methods we used, our results, and discussions on their wider implications in development of climate frameworks to meet energy targets.

Data

Using data from 1992 – UNFCCC framework from Rio Conference Earth Summit (conceptual foundation for this framework) we focused on filtering the data to countries that displayed full data for the share of low-carbon energy consumption. We started with the OWiD (Our World in Data) as our overall broad dataset then five from the IEA(International Energy Agency): four were for each policy type (Carbon pricing, coal phasing, emissions standard and low carbon spending) with the fifth containing all the power policies to cross reference which countries they had collected data for. Additionally, for Q.1 we included nuclear as part of the renewable energy consumption as no carbon emissions are produced in the process; nuclear energy tends to be more manageable than greenhouse gas emissions and has (relatively) minimal impact on climate change apart from rare catastrophes like Chernobyl.

Q.2 required further filtering from the original dataset to remove duplicates and entries that had null values that would skew the data. Once this was done, the dataset was filtered to show the power policies, and an executive decision was made on the policies that seemed most relevant to carbon-shares. For example, spending in low-carbon, carbon prices and coal phases outs taken from the IEA due to their relevance. We made a function that counted the number of policies that had been implemented for each country and year - this simplistic measure is fine because this is a scoping study and this will only increase the rates of a false negative as opposed to a false positive which is a better priority for the purpose of finding a relevant avenue for future study. Finally, the dataset was also filtered to show GDP per capita. We chose per capita instead of just GDP as per capita accounts for the countries' relative population, making the comparisons between countries with varying populations easier.

Model

Research Question 1: Is there evidence that the Sustainable Development Goals (2015) accelerated the adoption of low-carbon energy globally?

As mentioned in 'Data', the first dataset we used contains country data for each year 1992-2024, alongside the countries low carbon emission energy share (Before modelling this variable is renamed to share for better readability in the formula) and the GDP per capita for that year. The dataset also contains the country code and the world region (according to the source, our world in data) this column was kept when we considered how we might want to explore continental trends but was ultimately left out of scope, currently that data is redundant and unused.
Our hypothesis suggests that there is statistical significance to the idea that the SDGs have accelerated the adoption of low-carbon energy globally:

H_0: There is no statistical significance influencing the adoption of low-carbon energy consumption based on the SDGs
H_1: There is statistical significance suggesting the implementation of SDGs have accelerated the adoption of low-carbon energy sources.

Once our data was cleaned and exported as CleanData.csv,
We then ran this function to calculate average pre & post 2015 movement in countries' energy shares, by using the diff() function to compare change in neighboring array values.

```
CarbShares['Change'] = CarbShares['PostAvg'] - CarbShares['PreAvg']
formula_string = "Change ~ 1"
```

With the above formula we only mapped one variable, Change, this intercept-only model does not take in any predictors, instead we model changes against a constant.

The model here shows some signs of distress. The Kurtosis score is an indicator that there are large outliers in the model that are dragging data and decreasing the reliability; our P-value is 0 and indicates significance. The weak

coefficient of .44 suggests that there is an accelerated growth in the average shares of a country post 2015 but it is not a strong correlation, the model 'knows' that there are unknown variables that are not being considered. We chose to move forward with research question 2, however if we had revisited the first model, we could have restructured our data and added Entity/Time effects to better isolate the variables we were interested in.

To address this Kurtosis score, we plotted and labeled residuals of the tests:

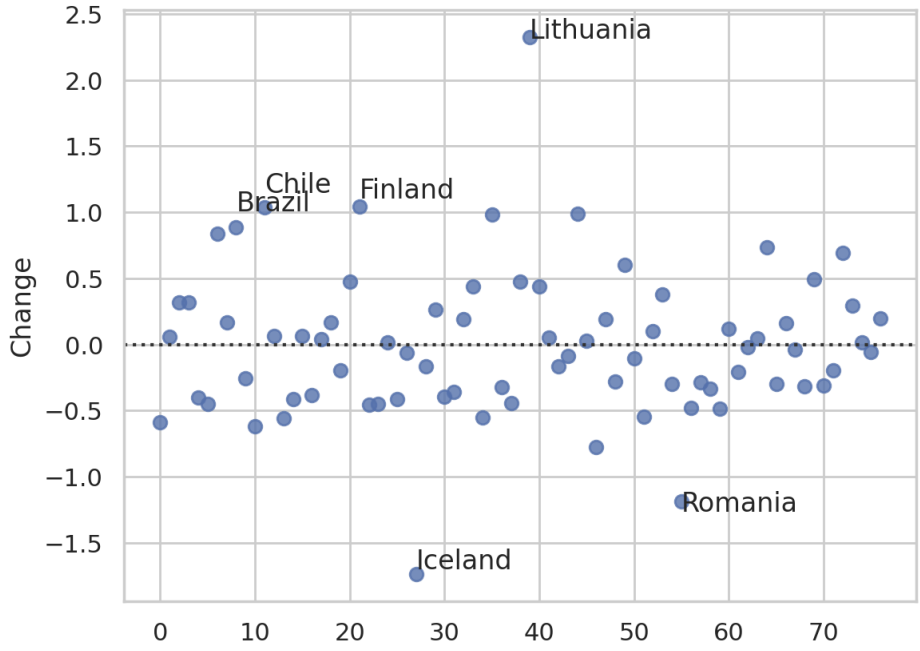

Figure 1.

This graph has the dotted line that represents the mean difference of 0.44, any countries that are more than 1 unit above or below have been labeled. A great expansion of this exercise would be to look at these outliers and the reasons why they may be so ahead or behind.
Our main takeaway from this graph was that we cannot compare countries against each other, and we kept this in mind when modeling research question 2.

Research Question 2: Does the presence of national energy policies influence the share of low-carbon energy consumption?

Our second question explores the potential effects that national energy policies have on low-carbon energy consumption. We explored this using the original dataset as provided by our world in data: 'Share of primary energy that is low-carbon vs. GDP per capita (2024)'. The variables prioritised to answer this question are the datasets describing the policies which are: carbon pricing, coal phasing, emissions standard and low carbon spending.

Our hypothesis is that the national energy policies had a positive influence on low-carbon energy consumption; the greater the number of policies, the greater the amount of success the country has in low-carbon energy consumption.

H_0: There is no statistical significance influencing the share of low-carbon energy consumption based on the number of national energy policies
H_1: There is statistical significance suggesting the number of national energy policies influence the share of low-carbon energy consumption

The model we used in Research Question 2 was also driven by our custom datasets, a collation of 5 sources:

The first dataset was our CleanData.csv from Q1, as well as these 4 CSVs
Together this data, once cleaned, was every Sustainable Energy Policy that aligned with our 77 countries in scope. From here we ran a function to tally each country's policy count over time; the output was clean data with no null values and ready to model.

Model 1:

A simple OLS model shows weak findings, suggesting that for every policy a country has there would be a ~2% increase of low carbon shares. This pooled model was giving false credit to policies when the real factors were time passing (Green energy is more accessible now that the 90s) and geography (Norway is suited for hydro power and so would logically have above-average shares). This autocorrelation is reflected in the Durbin-Watson of .06 suggesting strong positive correlation with error distribution. If policies were the main predictor, we would expect errors to fall above and below and the DW to sit between 1.5-2.5.

Model 2:

Here the Time (years) and Entity (country) are accounted for; the model is no longer confused by the time variables and knows to compare the countries independently This stops skewing due to global economic factors over time, or countries with natural (dis)advantages. Overall, this better isolates the effect of the policies. Specifically, a one-unit increase in the SDG policy count is associated with a 0.63 percentage point increase in the low-carbon energy share in the following year P = 0.009.

Note: this made the alleged impact smaller (~.5%), but the certainty higher, an indicator that we have improved the model's reliability.

Model 3:

We can expect a policy passed to have an impact, but logically we think that this effect will be delayed as adoption takes time. With a time lag, we can model the policy impact 2 years after implementation.
We also modeled with a 1-year time lag but found that confidence was stronger in the second model whilst 1 year sat between 2 years and no lag.

Now we have better isolated our predictor, we can see the P-value shows our statistical significance. With a coefficient of ~.93 there is a near 1% increase in a country's low carbon emission share per policy.
The Entity and Time effects have impacted the R-Squared values, which is logical when we are assessing policy impacts across many countries and years.

Model 4:

Now we will control GDP per capita. This is to check if policies are exclusively done by richer countries as they can afford to or if the policies are the true predictors of share variation across all countries.

The policy coefficient remains high; the GDP per capita wasn't a great indication of shares, so we can say that the policy, more than the wealth, is the better predictor.

Results

Q1:
The results of the linear regression analysis showed brought about a P-value of <0.05 meaning we can reject the null hypothesis. The test calculated the average share of LCE per country pre and post 2015. Figure one shows the average distance from the mean is 0.44. The coefficient pre 2015 is 0.4464, and 0.8964 post 2015 which suggests a strong positive correlation between SDG and adoption of LCE. Therefore, countries cannot be compared with each other as there are other factors which are impacting the adoption of LCE in any given country.

Q2:
The results of the t-test show that the resulting P-value is 0.0044 < 0.05 which means we can reject H_0, proving there is statistical significance that the introduction of national energy policies has positively influenced countries' the share of low-carbon energy consumption. Additionally, we ran a test regarding the GDP for each country as some countries have a higher GDP that could influence the low-carbon energy consumption. The results showed that the p-value was 0.1357 > 0.05 suggesting there is no statistical significance, proving that the main effect of low-carbon consumption is due to the policies not the GDP of the country.

Discussion

This paper investigated two research questions: whether the introduction of SDG 7 in 2015 accelerated the global adoption of low-carbon energy, and whether national energy policy commitments influence countries' low-carbon energy shares. Using data from 77 countries between 1992 and 2024, we analysed changes in low-carbon energy shares before and after 2015 and assessed the relationship between policy counts and low-carbon energy adoption using linear regression and panel fixed-effects models whilst controlling for GDP per capita.

SDG 7 and Global Acceleration of Low-Carbon Energy

Our first research question explored whether SDG 7 corresponded with a measurable acceleration in low-carbon energy adoption. We calculated the average annual change in low-carbon energy share for each country before and after 2015, using the difference between post-2015 and pre-2015 averages as our acceleration measure. The results show that the mean increase in yearly low-carbon energy share after 2015 was 0.45 percentage points higher than before 2015, with a p-value below 0.001. This allows us to reject the null hypothesis and suggests that the introduction of SDG 7 is associated with a statistically significant increase in the global shift towards low-carbon energy adoption.

However, the plotted data shows considerable heterogeneity. Countries such as Chile, Brazil and Lithuania show particularly strong post-2015 growth, while others exhibit minimal change or even regression, as in the cases of Iceland and Romania. This uneven response aligns with existing literature, emphasising that renewable energy adoption is shaped by national policy frameworks, resource availability, investment capacity and political commitment. While SDG 7 appears to have acted as a global catalyst, its effects are not homogeneous, clearly mediated by national contexts.

To explore whether economic development explains this variation, we weighted the model by GDP per capita. The coefficient increased slightly from 0.45 to 0.462, but the difference is negligible. Whilst this confirms that higher-income countries may experience marginally faster growth, GDP per capita is not the primary driver of post-2015 acceleration – we see that acceleration apparently related to the SDGs not to be limited to wealthier countries. This finding anticipates the results of our second research question, suggesting that policy commitment may matter more than economic output.

National Energy Policies and Low-Carbon Energy Shares

Our second research question explores the potential effects that national energy policies have on low-carbon energy consumption. The variables prioritised to answer this question were: carbon pricing, coal phasing, emissions standard and low carbon spending. With this, we created a policy count variable and combined it with the low-carbon energy share data.

To calculate the statistical significance, the original dataset was first adjusted to show only the countries that have complete data for both policy and low-carbon share, and then pre- and post-2015 averages were calculated. The statistical tests conducted assessed the relationship between policy count and low-carbon energy shares.

Limitations

Several limitations should be acknowledged. First, the OWID dataset contains inconsistencies across countries and years, and the need to exclude countries with incomplete data may bias results towards more stable, higher-income countries. Second, our acceleration measure is based on average changes before and after 2015, which may obscure year-to-year volatility, particularly in lower-income or politically unstable countries. Third, the one-sample test used in the SDG 7 analysis cannot identify causal mechanisms, and the model does not account for the panel structure of the data. Fourth, policy counts treat all policies as equal, despite differences in scale, duration, jurisdiction and effectiveness. Finally, outliers can influence mean changes, and our models do not fully capture the complexity of national energy transitions.

Future Research

Future work could employ panel regression models to more rigorously test the causal effects of SDG 7 and national policies, accounting for time-varying factors and country-specific characteristics. Further research could also develop a more nuanced policy variable that incorporates policy strength, financial scale, jurisdictional reach and time lags. Clustering countries by income level, political stability or energy mix could help explain heterogeneous responses to SDG 7. Sector-specific analyses may reveal whether acceleration is concentrated in particular areas of energy production. Finally, validating these findings using alternative datasets or extending the analysis into the 2020s would help determine whether the post-2015 acceleration persists or plateaus.

References

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